Schemes Gently: A Guided Tour of Spectrum, Nilpotents, and Local Rings

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§1.1 Prime ideals, generic points, and specialization

The set of solutions of a polynomial equation does not, by itself, remember generic subvarieties or infinitesimal structure. Following the construction of affine schemes in the *Stacks Project*, we examine how a ring determines a topological space together with local rings.[1]

For a ring R, define

$$\operatorname{Spec} R = \{\mathfrak{p} \subset R : \mathfrak{p} \text{ is prime}\}.$$

This is the spectrum of R. Its points are prime ideals.

At first this feels like a category mistake: ideals are algebraic objects, not points. But this change is exactly what allows geometry to remember generic behavior, arithmetic information, and infinitesimal thickening.

For $R = k[x]$ over an algebraically closed field k, the maximal ideals

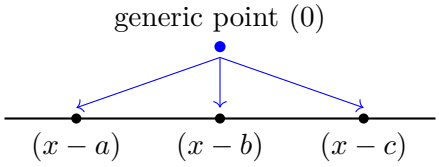
$$(x - a), \qquad a \in k,$$

correspond to ordinary closed points. This requires the algebraically closed hypothesis: over $\mathbb{R}$, the ideal $(x^2 + 1)$ is also maximal, and its residue field is $\mathbb{C}$. There is also the prime (0), the generic point.

The closure of the generic point is the whole space:

$$\overline{\{(0)\}} = \operatorname{Spec} k[x].$$

This is not Hausdorff behavior, but it is meaningful. The generic point represents the whole irreducible line, while $(x - a)$ represents a closed point.



The arrows are suggestive: the generic point specializes to closed points.

It is tempting to keep only maximal ideals, because over an algebraically closed field they correspond to ordinary closed points. But prime ideals contain information about irreducible subspaces.

In $k[x, y]$, the prime ideal (x) is not a closed point. It corresponds to the whole y-axis. The maximal ideal $(x, y - a)$ is a closed point on that axis.

Thus prime ideals let a scheme contain both points and subvarieties as points of different levels of generality. This is strange at first, but it is exactly what makes generic points possible.

[1] Stacks Project, *The spectrum of a ring* (Tag 00DY), and *Affine schemes* (Tag 01HR).

In $\operatorname{Spec} R$, one says that $\mathfrak{q}$ specializes to $\mathfrak{p}$ when

$$\mathfrak{q} \subseteq \mathfrak{p}.$$

The smaller prime is more generic; the larger prime is more special.

For $k[x]$,

$$(0) \subset (x-a),$$

so the generic point specializes to the closed point $(x-a)$. This order relation is the topological shadow of ideal containment.

For an ideal $I \subset R$, define

$$V(I) = \{\mathfrak{p} \in \operatorname{Spec} R : I \subseteq \mathfrak{p}\}.$$

These are closed sets.

For $f \in R$, define

$$D(f) = \{\mathfrak{p} : f \notin \mathfrak{p}\}.$$

This is the region where f is allowed to be inverted. Distinguished opens are the basic open sets of affine schemes.

§1.2 Functions near a point

At a prime ideal $\mathfrak{p}$, the local ring is

$$R_{\mathfrak{p}}.$$

It is obtained by inverting every element not in $\mathfrak{p}$. The interpretation is simple: near $\mathfrak{p}$, functions not vanishing there should be units.

For $R = k[x]$ and $\mathfrak{p} = (x-a)$, the ring

$$k[x]_{(x-a)}$$

contains rational functions whose denominators do not vanish at a. This is the algebraic version of functions defined near a.

A spectrum is not only a topological space. It carries functions. On a distinguished open $D(f)$, the structure sheaf satisfies

$$\mathcal{O}_{\operatorname{Spec} R}(D(f)) = R_f.$$

The stalk at $\mathfrak{p}$ is

$$\mathcal{O}_{\operatorname{Spec} R,\mathfrak{p}} = R_{\mathfrak{p}}.$$

These two formulas make local calculations possible; the sheaf condition supplies the missing compatibility. If $D(f) = \bigcup_i D(f_i)$, sections from the rings R_{f_i} glue to a unique section of R_f precisely when they agree after restricting to the overlaps $R_{f_i f_j}$. This gives the algebraic exactness pattern $0 \to R_f \to \bigoplus_i R_{f_i} \to \bigoplus_{i,j} R_{f_i f_j}$, where the final map records pairwise differences.

At a prime $\mathfrak{p}$, the local ring $R_{\mathfrak{p}}$ has maximal ideal $\mathfrak{p}R_{\mathfrak{p}}$. The residue field is

$$\kappa(\mathfrak{p}) = R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}}.$$

For a closed point $(x-a) \subset k[x]$, the residue field is k. For the generic point (0), the residue field is $k(x)$, the field of rational functions.

This is another way to see the meaning of generic points: at the generic point of the affine line, the coordinate x is not specialized to a number; it remains a variable.

§1.3 Arithmetic and local prime information

The spectrum of the integers has points

$$(p)$$

for prime numbers p, together with the generic point (0). An integer n vanishes at exactly the primes dividing it:

$$V(n) = \{(p) : p \mid n\}.$$

Thus divisibility becomes geometry.

This converts divisibility into a concrete local calculation. On $D(6)$, the primes (2) and (3) have been removed, and the ring of regular sections is $\mathbb{Z}[1/6]$. At (p), the local ring is $\mathbb{Z}_{(p)}$, whose denominators are exactly the integers not divisible by p. Its residue field is $\mathbb{F}_p$, while the generic point (0) has residue field $\mathbb{Q}$.

§1.4 Invisible Infinitesimal Structure

Consider

$$R = k[x]/(x^2).$$

The element x is nonzero but satisfies $x^2 = 0$. Topologically, $\operatorname{Spec} R$ has only one point. Algebraically, it remembers an infinitesimal thickening of that point.

A useful picture is a point with a first-order tangent fuzz. The fuzz cannot be seen as a second classical point, but it can be detected by functions.

This is one of the reasons schemes are powerful: they keep information that ordinary point sets discard.

Let

$$R = k[\epsilon]/(\epsilon^2).$$

A function on $\operatorname{Spec} R$ can have the form

$$a + b\epsilon.$$

The term $b\epsilon$ is invisible if one only looks at ordinary points, because ϵ is nilpotent. But algebraically it records first-order infinitesimal information.

The dual-number model detects a derivative directly. For a polynomial $f(x)$ and an element $a + b\epsilon$, the binomial theorem gives

$$f(a + b\epsilon) = f(a) + bf'(a)\epsilon$$

because all terms with ϵ^2 vanish. The coefficient b is a first-order tangent direction; ordinary points see only the value $f(a)$. This is a calculation, rather than a drawing of a point with vaguely described “fuzz”.